

Breaking Bob's Weak RSA Key

Public Key: $(n, e) = (3139, 883)$

Introduction

Bob's RSA key is weak because the modulus n is small. To recover his private key d , we need to:

1. Factor n into primes p and q .
2. Compute Euler's totient $\varphi(n) = (p - 1)(q - 1)$.
3. Compute the modular inverse $d = e^{-1} \bmod \varphi(n)$ using the Extended Euclidean Algorithm.

—

Step 1: Factor n

Bob's modulus is:

$$n = 3139$$

We only need to test prime numbers up to $\sqrt{3139} \approx 56$.

Primes up to 56:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53$$

We divide 3139 by each in turn:

$$3139/2 = 1569.5 \quad (\text{not integer})$$

$$3139/3 = 1046.33 \dots \quad (\text{not integer})$$

$$\vdots$$

$$3139/43 = 73 \quad (\text{integer})$$

Step-by-step verification of multiplication:

$$43 \times 73 = 43 \times (70 + 3) = 3010 + 129 = 3139$$

Hence, the prime factors are:

$$p = 43, \quad q = 73$$

—

Step 2: Compute Euler's Totient

$$\varphi(n) = (p - 1)(q - 1) = 42 \cdot 72 = 3024$$

—

Step 3: Compute the Private Key d

We need d such that:

$$d \cdot e \equiv 1 \pmod{\varphi(n)}, \quad \text{i.e., } 883d \equiv 1 \pmod{3024}$$

—

Step 3a: Euclidean Algorithm

Compute $\gcd(3024, 883)$:

$$3024 = 883 \cdot 3 + 375$$

$$883 = 375 \cdot 2 + 133$$

$$375 = 133 \cdot 2 + 109$$

$$133 = 109 \cdot 1 + 24$$

$$109 = 24 \cdot 4 + 13$$

$$24 = 13 \cdot 1 + 11$$

$$13 = 11 \cdot 1 + 2$$

$$11 = 2 \cdot 5 + 1$$

$$2 = 1 \cdot 2 + 0$$

The last nonzero remainder is 1, so $\gcd(883, 3024) = 1$, confirming the modular inverse exists.

—

Step 3b: Extended Euclidean Algorithm (Full Detailed Version)

We want integers d and k such that:

$$883 \cdot d + 3024 \cdot k = 1$$

We now express 1 as a linear combination of 883 and 3024 by back-substitution. Start from the last nonzero remainder:

$$1 = 11 - 2 \cdot 5$$

Substitute $2 = 13 - 11$:

$$\begin{aligned} 1 &= 11 - 5(13 - 11) \\ &= 11 - 5 \cdot 13 + 5 \cdot 11 \\ &= 6 \cdot 11 - 5 \cdot 13 \end{aligned}$$

Substitute $11 = 24 - 13$:

$$\begin{aligned} 1 &= 6(24 - 13) - 5 \cdot 13 \\ &= 6 \cdot 24 - 6 \cdot 13 - 5 \cdot 13 \\ &= 6 \cdot 24 - 11 \cdot 13 \end{aligned}$$

Substitute $13 = 109 - 24 \cdot 4$:

$$\begin{aligned} 1 &= 6 \cdot 24 - 11(109 - 4 \cdot 24) \\ &= 6 \cdot 24 - 11 \cdot 109 + 44 \cdot 24 \\ &= 50 \cdot 24 - 11 \cdot 109 \end{aligned}$$

Substitute $24 = 133 - 109$:

$$\begin{aligned} 1 &= 50(133 - 109) - 11 \cdot 109 \\ &= 50 \cdot 133 - 50 \cdot 109 - 11 \cdot 109 \\ &= 50 \cdot 133 - 61 \cdot 109 \end{aligned}$$

Substitute $109 = 375 - 2 \cdot 133$:

$$\begin{aligned} 1 &= 50 \cdot 133 - 61(375 - 2 \cdot 133) \\ &= 50 \cdot 133 - 61 \cdot 375 + 122 \cdot 133 \\ &= 172 \cdot 133 - 61 \cdot 375 \end{aligned}$$

Substitute $133 = 883 - 2 \cdot 375$:

$$\begin{aligned} 1 &= 172(883 - 2 \cdot 375) - 61 \cdot 375 \\ &= 172 \cdot 883 - 344 \cdot 375 - 61 \cdot 375 \\ &= 172 \cdot 883 - 405 \cdot 375 \end{aligned}$$

Substitute $375 = 3024 - 3 \cdot 883$:

$$\begin{aligned} 1 &= 172 \cdot 883 - 405(3024 - 3 \cdot 883) \\ &= 172 \cdot 883 - 405 \cdot 3024 + 1215 \cdot 883 \\ &= (172 + 1215) \cdot 883 - 405 \cdot 3024 \\ &= 1387 \cdot 883 - 405 \cdot 3024 \end{aligned}$$

Thus we have:

$$1 = 1387 \cdot 883 - 405 \cdot 3024$$

which shows that the modular inverse of 883 (mod 3024) is

$$d = 1387.$$

—

Step 3c: Coefficient Table for the Extended Euclidean Algorithm

To make the process even clearer, we track each remainder r_i as a linear combination

$$r_i = a_i \cdot 883 + b_i \cdot 3024.$$

i	r_i	r_i as $a_i \cdot 883 + b_i \cdot 3024$
0	$r_0 = 3024$	$0 \cdot 883 + 1 \cdot 3024$
1	$r_1 = 883$	$1 \cdot 883 + 0 \cdot 3024$
2	$r_2 = 375$	$-3 \cdot 883 + 1 \cdot 3024$
3	$r_3 = 133$	$7 \cdot 883 - 2 \cdot 3024$
4	$r_4 = 109$	$-17 \cdot 883 + 5 \cdot 3024$
5	$r_5 = 24$	$24 \cdot 883 - 7 \cdot 3024$
6	$r_6 = 13$	$-113 \cdot 883 + 33 \cdot 3024$
7	$r_7 = 11$	$137 \cdot 883 - 40 \cdot 3024$
8	$r_8 = 2$	$-250 \cdot 883 + 73 \cdot 3024$
9	$r_9 = 1$	$1387 \cdot 883 - 405 \cdot 3024$

Note that the final row corresponds to:

$$1 = 1387 \cdot 883 - 405 \cdot 3024,$$

which again confirms that

$$d = 1387$$

is the modular inverse of 883 modulo 3024.

—

Step 4: Bob's Private Key

Bob's private key is:

$$d = 1387$$

Full key pair:

$$(n, e, d) = (3139, 883, 1387)$$

—

Conclusion

Bob's RSA key is weak because:

- The modulus n is small enough to factor by hand.
- Once p and q are known, computing d via the Extended Euclidean Algorithm is straightforward.

Decrypting Alice's Message with Bob's RSA Key

Public Key: $(n, e) = (3139, 883)$

Private Key: $d = 1387$

Problem

Bob uses RSA with public key

$$(n, e) = (3139, 883).$$

We have already found that his private key is

$$d = 1387.$$

Alice encrypts a message using Bob's public key, and the encrypted value (ciphertext) is

$$C = 3121.$$

Our task is to find the original plaintext message m .

In RSA decryption, we compute

$$m \equiv C^d \pmod{n}.$$

So here we want

$$m \equiv 3121^{1387} \pmod{3139}.$$

Directly computing 3121^{1387} is impossible by hand, so we use modular arithmetic with the prime factors of n .

Step 1: Use that $n = 43 \cdot 73$

From earlier work we know:

$$n = 3139 = 43 \cdot 73.$$

We will:

1. First compute $m \bmod 43$.
2. Then compute $m \bmod 73$.
3. Finally find the unique number m (with $0 \leq m < 3139$) that satisfies both congruences.

This avoids huge powers and uses only smaller numbers.

Step 2: Work modulo 43

First reduce C modulo 43:

$$3121 = 43 \cdot 72 + 25 \quad \Rightarrow \quad 3121 \equiv 25 \pmod{43}.$$

So we need

$$m \equiv 25^{1387} \pmod{43}.$$

Because 43 is prime, Fermat's little theorem tells us:

$$a^{42} \equiv 1 \pmod{43} \quad \text{for all } a \text{ not divisible by } 43.$$

Here $\gcd(25, 43) = 1$, so

$$25^{42} \equiv 1 \pmod{43}.$$

Now reduce the exponent 1387 modulo 42:

$$1387 = 42 \cdot 33 + 1 \quad \Rightarrow \quad 1387 \equiv 1 \pmod{42}.$$

Thus:

$$25^{1387} = 25^{42 \cdot 33 + 1} = (25^{42})^{33} \cdot 25^1 \equiv 1^{33} \cdot 25 \equiv 25 \pmod{43}.$$

So we have:

$$m \equiv 25 \pmod{43}.$$

Step 3: Work modulo 73

Now reduce C modulo 73:

$$3121 = 73 \cdot 42 + 55 \Rightarrow 3121 \equiv 55 \pmod{73}.$$

So we need

$$m \equiv 55^{1387} \pmod{73}.$$

Again, 73 is prime, so Fermat's little theorem gives:

$$a^{72} \equiv 1 \pmod{73} \text{ for all } a \text{ not divisible by } 73.$$

Since $\gcd(55, 73) = 1$, we have:

$$55^{72} \equiv 1 \pmod{73}.$$

Now reduce the exponent 1387 modulo 72:

$$1387 = 72 \cdot 19 + 19 \Rightarrow 1387 \equiv 19 \pmod{72}.$$

Therefore:

$$55^{1387} \equiv 55^{19} \pmod{73}.$$

We now compute $55^{19} \pmod{73}$ using repeated squaring.

Compute powers of 55 modulo 73

$$55^1 \equiv 55 \pmod{73},$$

$$55^2 = 55 \cdot 55 = 3025,$$

$$3025 \equiv 3025 - 73 \cdot 41 = 3025 - 2993 = 32 \pmod{73},$$

$$55^4 \equiv (55^2)^2 \equiv 32^2 = 1024 \equiv 1024 - 73 \cdot 14 = 1024 - 1022 = 2 \pmod{73},$$

$$55^8 \equiv (55^4)^2 \equiv 2^2 = 4 \pmod{73},$$

$$55^{16} \equiv (55^8)^2 \equiv 4^2 = 16 \pmod{73}.$$

Now we write:

$$19 = 16 + 2 + 1,$$

so

$$55^{19} = 55^{16} \cdot 55^2 \cdot 55^1.$$

Substitute the values:

$$55^{19} \equiv 16 \cdot 32 \cdot 55 \pmod{73}.$$

Compute step by step:

$$16 \cdot 32 = 512.$$

Reduce 512 modulo 73:

$$73 \cdot 7 = 511, \quad 512 - 511 = 1 \Rightarrow 512 \equiv 1 \pmod{73}.$$

Thus:

$$55^{19} \equiv 1 \cdot 55 \equiv 55 \pmod{73}.$$

So we have:

$$m \equiv 55 \pmod{73}.$$

Step 4: Solve the two congruences

We now know:

$$\begin{cases} m \equiv 25 \pmod{43}, \\ m \equiv 55 \pmod{73}. \end{cases}$$

We solve this system without naming any theorems; just use algebra.

Write:

$$m = 25 + 43k \quad \text{for some integer } k.$$

Plug this into the second congruence:

$$25 + 43k \equiv 55 \pmod{73}.$$

Subtract 25 from both sides:

$$43k \equiv 30 \pmod{73}.$$

Now we need the inverse of 43 modulo 73, i.e., a number t such that:

$$43t \equiv 1 \pmod{73}.$$

Try $t = 17$:

$$43 \cdot 17 = 731 \equiv 731 - 73 \cdot 10 = 731 - 730 = 1 \pmod{73}.$$

So $43^{-1} \equiv 17 \pmod{73}$.

Multiply both sides of $43k \equiv 30 \pmod{73}$ by 17:

$$k \equiv 17 \cdot 30 \pmod{73}.$$

Compute:

$$17 \cdot 30 = 510.$$

Reduce 510 modulo 73:

$$73 \cdot 6 = 438, \quad 510 - 438 = 72,$$

so

$$k \equiv 72 \pmod{73}.$$

Now plug back into $m = 25 + 43k$ with $k = 72$:

$$m = 25 + 43 \cdot 72.$$

Compute:

$$43 \cdot 72 = 43(70 + 2) = 43 \cdot 70 + 43 \cdot 2 = 3010 + 86 = 3096.$$

So:

$$m = 25 + 3096 = 3121.$$

Since $0 \leq 3121 < 3139$, this is the unique solution modulo 3139.

$$\boxed{m = 3121}$$

Conclusion

The decrypted plaintext message corresponding to the ciphertext $C = 3121$ is

$$\boxed{3121}.$$

In this particular case, the ciphertext and plaintext have the same numerical value when interpreted modulo $n = 3139$.